



JOHNS HOPKINS

WHITING SCHOOL
of ENGINEERING

FPGA Design Using VHDL

Module 1B: FPGA Overview

Module Overview



- Recall how FPGAs operate in general
- Describe combinational logic using VHDL
- Compare different styles of combinational logic
- Construct a simple design using combinational logic

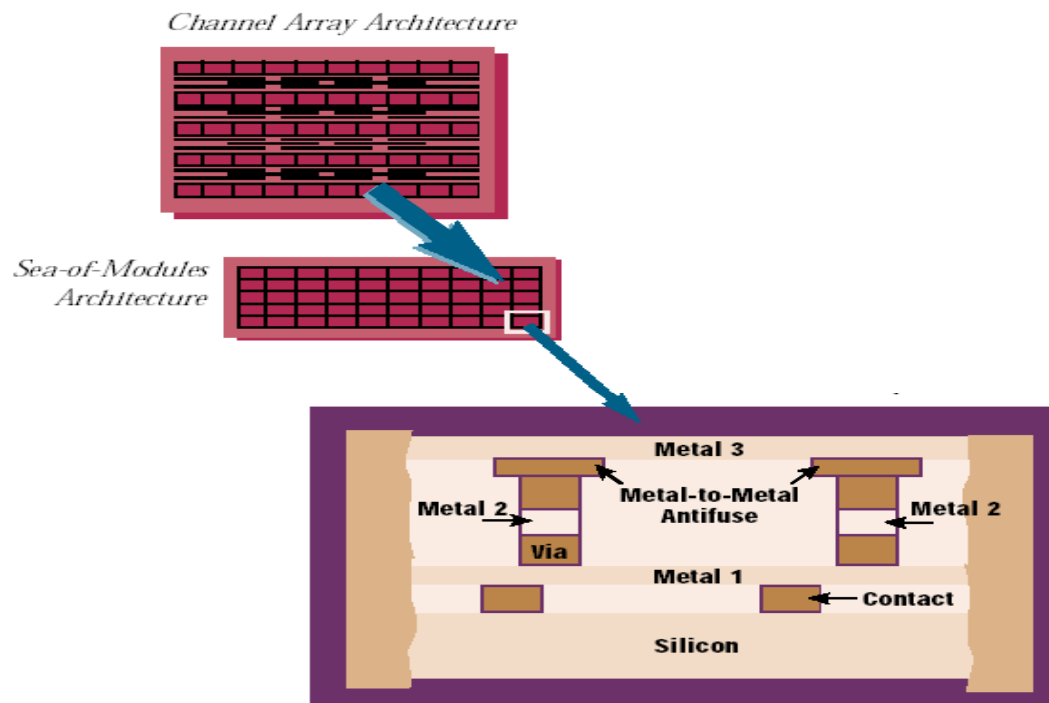
Digital System Considerations

	ASIC	General Processor	FPGA
Speed	FAST	Slow	FAST
Flexibility	None	Very Flexible	Limited
Time to Market	Long	Immediate	Immediate
Production Cost	Low (after NRE)	Expensive	Moderate

- Implemented on hardware with the flexibility and ease of changing as software

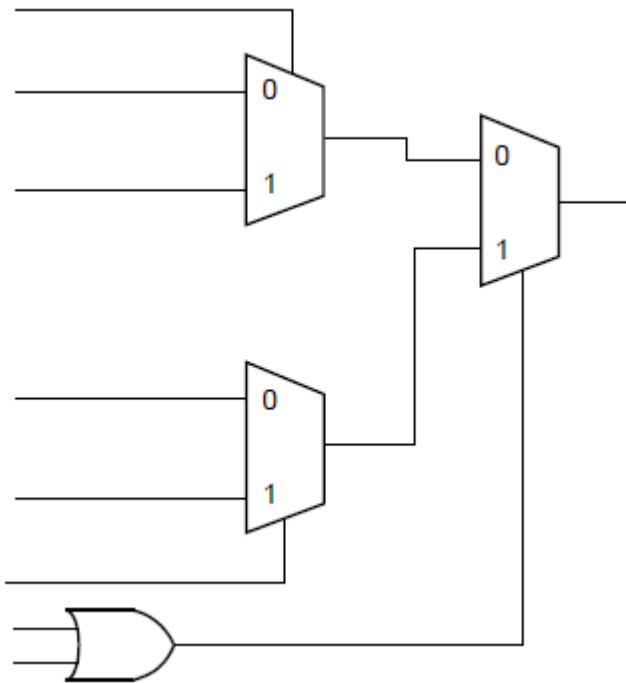
“Traditional” FPGA Architecture

- Modules are connected by “Programmable” link
- One-time Programmable Only

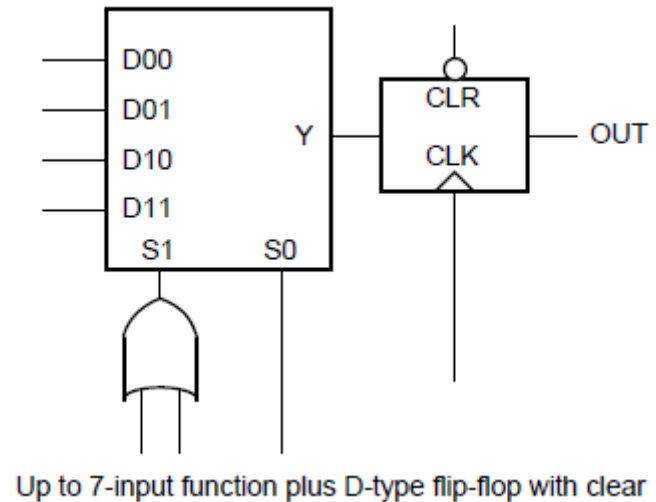


Microsemi Logic Module

- Combinational Logic Module



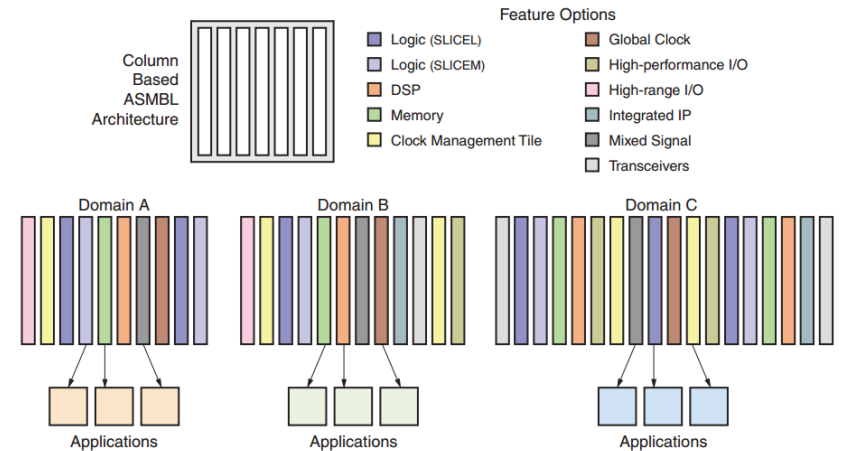
- Sequential Logic Module



Note: Both figures obtained from Microsemi Actel App Note AC165

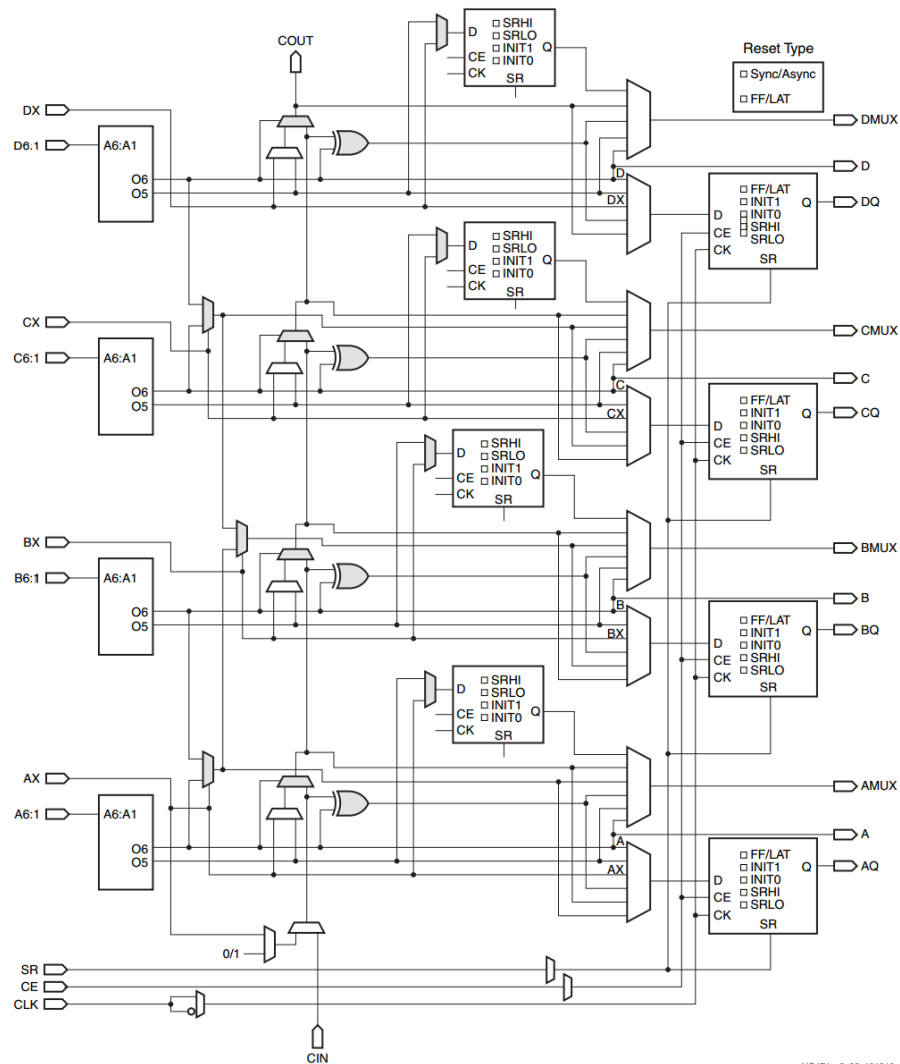
Xilinx Artix-7 Architecture

- 7 Series FPGA Architecture from Xilinx
- Configurable Logic Blocks (CLB)
 - Built up of slices (usually a pair)
 - SLICEL and SLICEM
 - Contain combinatorial and sequential circuits
- Slices are configurable



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Xilinx Artix-7 Architecture Continued

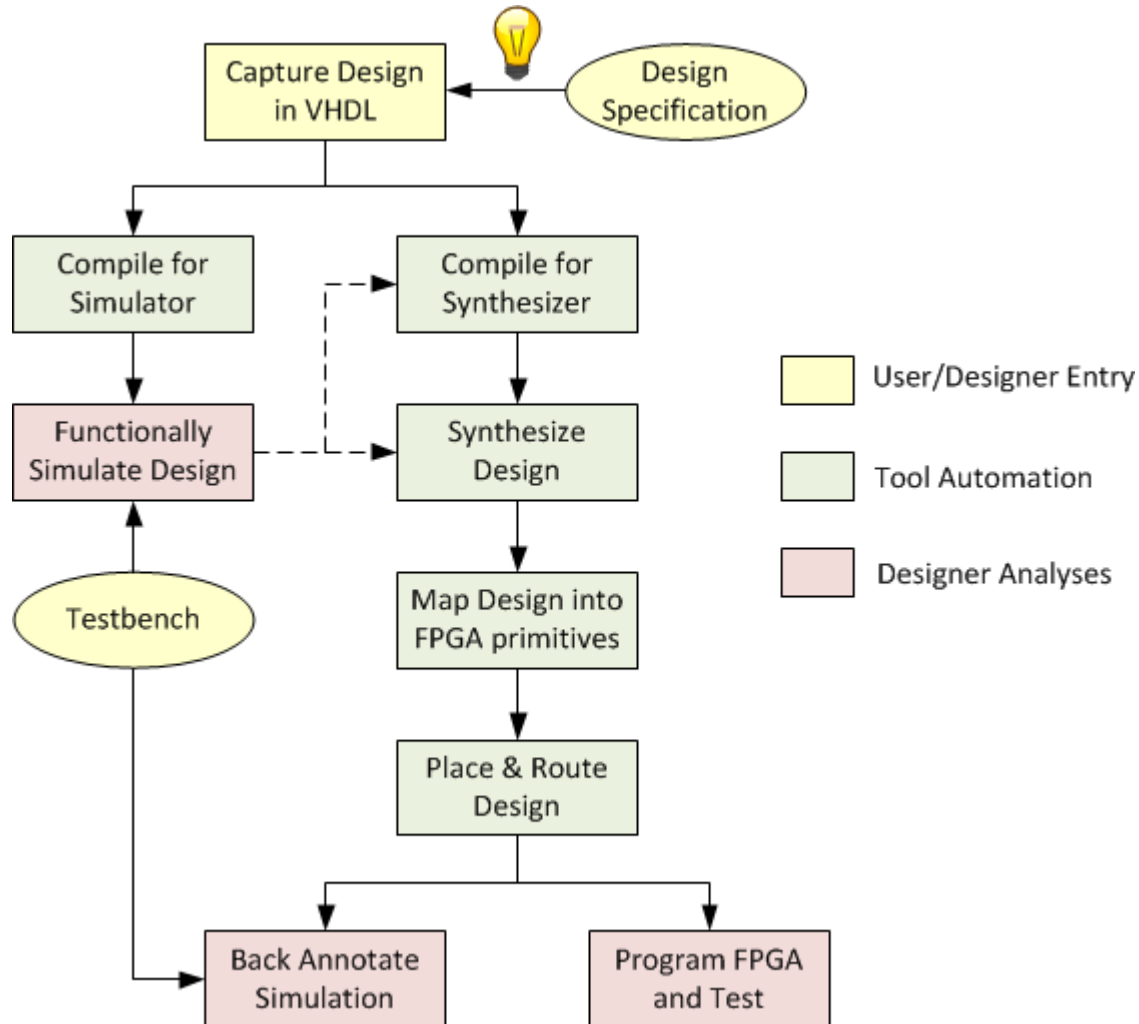


Design Flow



- Capture Design
- Compile
- Functional Simulation
- Synthesis/Map
- Place and Route
- Back Annotate/Timing Simulation
- Bitstream Generation and Program

Design Flow Continued



More FPGA History



- For more information on the history of FPGAs and how they compare to ASICs and other design techniques, please take a look at the optional reading assignments